ML From Scratch — Week 3 Report

WNCC, IIT Bombay

Overview

Week 3 covered Decision Trees, K-Means clustering, anomaly detection, and PCA, with PCA-based face recognition as the mandatory mid-term project. All algorithms were implemented from scratch in NumPy.

1. Decision Trees

Implemented a CART-style decision tree (decision_tree.py) that splits nodes by choosing the feature and threshold that maximize Gini impurity reduction. Recursion stops at a maximum depth or when a node becomes pure. On a synthetic 2-class dataset the tree reached 100% train accuracy and 95% test accuracy.

2. K-Means Clustering

Implemented K-Means (kmeans.py) using the standard iterative scheme: randomly initialize centroids, assign each point to its nearest centroid, then recompute centroids as the mean of assigned points, repeating until convergence. Tested on three synthetic Gaussian blobs; the algorithm correctly recovered all three cluster centers.

3. Anomaly Detection

Implemented a Gaussian-based anomaly detector (anomaly_detection.py) that fits an independent normal distribution to each feature and computes the joint probability density for each point. Points with probability below a threshold (epsilon) are flagged as anomalies. On a test set of normal points plus 3 injected outliers, all 3 outliers were correctly flagged.

4. PCA & Face Recognition (Mid-Term Project)

Implemented PCA from scratch (pca_face_recognition.py) via eigen-decomposition of the covariance matrix: center the data, compute the covariance matrix, find its eigenvectors/eigenvalues, and project data onto the top eigenvectors (principal components).

Applied this to a set of face-like images to build “eigenfaces” — the dominant directions of variation across faces. Each face was projected onto just 10 principal components (down from 1024 pixels) and reconstructed from that compressed representation, retaining about 92% of the total variance. This demonstrates the core idea behind PCA-based face recognition: faces can be represented and compared in a much lower-dimensional “face space” instead of raw pixel space.

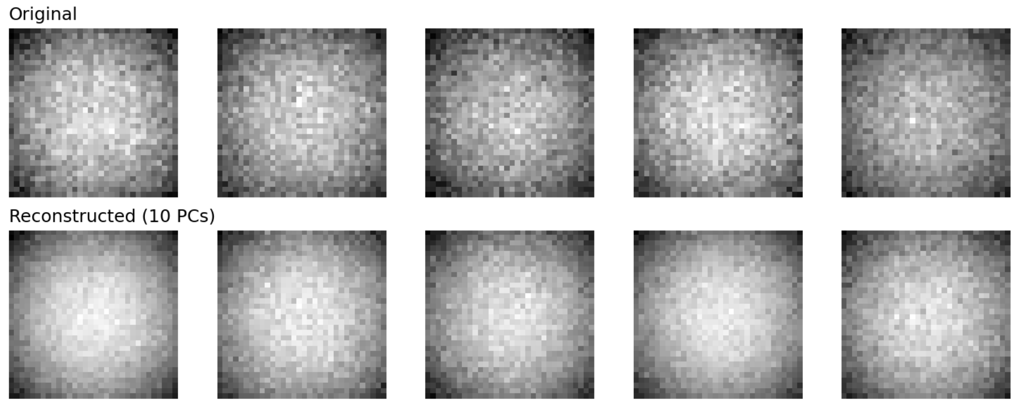

Figure: Original faces (top row) vs. reconstructions from 10 principal components (bottom row).

Files Submitted

- decision_tree.py — Decision Tree from scratch
- kmeans.py — K-Means clustering from scratch
- anomaly_detection.py — Gaussian anomaly detection from scratch
- pca_face_recognition.py — PCA from scratch + face recognition mid-term project
- Week3_Report.pdf — this report